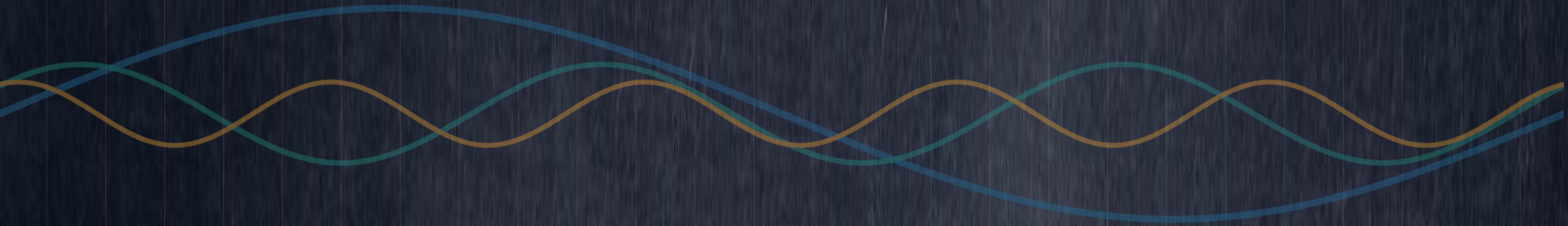


FOURIER INTEGRALS & TRANSFORMS

Arshad Afzal, PhD

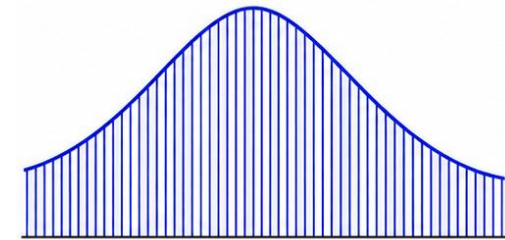
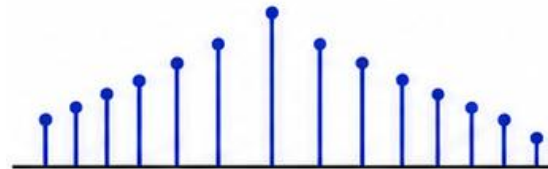
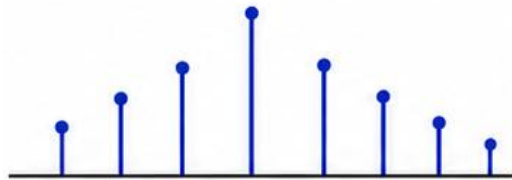
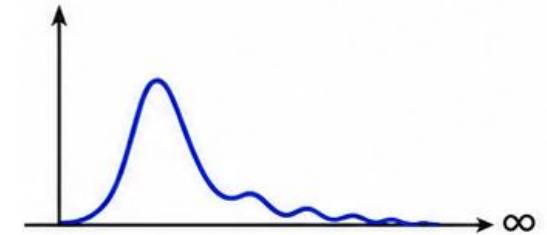
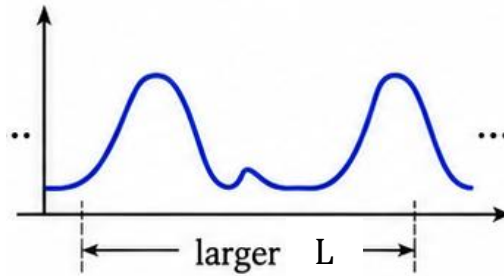
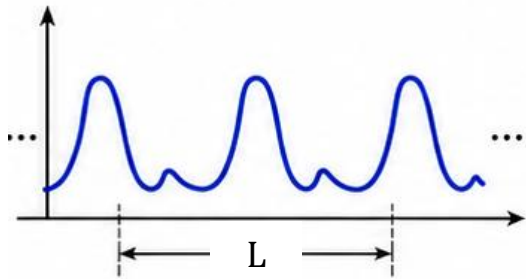


Fourier Integral

» Fourier integral

- Modeling non – periodic functions,

$$\text{As } L \rightarrow \infty; \quad f(x) = \lim_{L \rightarrow \infty} f_L(x)$$



Discrete harmonics, nw_0 turn into continuous variable, w and the coefficients, c_n turn into a function, $f(w)$

Fourier Integral

» Fourier integral

- Modeling non – periodic functions on the whole x – axis using extension of method of Fourier series
- If $f(x)$ is piecewise continuous in every finite interval, and has right-hand derivative and left-hand derivative at every point , and if below integral exists series

$$\lim_{a \rightarrow -\infty} \int_a^0 |f(x)| dx + \lim_{b \rightarrow \infty} \int_0^b |f(x)| dx$$

$$f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw$$

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv$$

- For points where $f(x)$ is discontinuous (say, x_0) , the sum of series is the average of the left- and right-hand limits of $f(x)$ at x_0

Fourier Integral

» Fourier integral

- For Consider any periodic function $f_L(x)$ of period $2L$ that can be represented by a Fourier series

$$f_L(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos w_n x + b_n \sin w_n x, \quad w_n = \frac{n\pi}{L}$$

When $L \rightarrow \infty$; $f(x) = \lim_{L \rightarrow \infty} f_L(x)$

$$f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{L} \sum_{n=1}^{\infty} \left[\cos w_n x \int_{-L}^L f_L(v) \cos w_n v dv + \sin w_n x \int_{-L}^L f_L(v) \sin w_n v dv \right]$$

$$\Delta w = w_{n+1} - w_n = \frac{(n+1)\pi}{L} - \frac{n\pi}{L} = \frac{\pi}{L}$$

$$\frac{1}{L} = \frac{\Delta w}{\pi}$$

$$f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{\pi} \sum_{n=1}^{\infty} \left[\cos w_n x \Delta w \int_{-L}^L f_L(v) \cos w_n v dv + \sin w_n x \Delta w \int_{-L}^L f_L(v) \sin w_n v dv \right]$$

As $L \rightarrow \infty$ $\frac{1}{2L} \int_{-L}^L f_L(v) dv = 0$

Fourier Integral

Also, as $\Delta\omega = \frac{\pi}{L} \rightarrow 0$ as $L \rightarrow \infty$, the infinity series becomes an integral from 0 to ∞

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\cos wx \int_{-\infty}^{\infty} f(v) \cos wv dv + \sin wx \int_{-\infty}^{\infty} f(v) \sin wv dv \right] dw$$

If we introduce the notation

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv$$

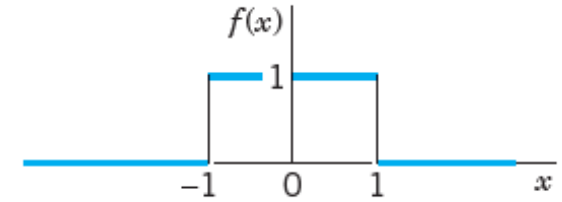
$$f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw$$

This is called the representation of $f(x)$ by a Fourier integral.

Fourier Integral

» Example

$$f(x) = \begin{cases} 1, & \text{if } |x| < 1 \\ 0, & \text{if } |x| > 1 \end{cases}$$



Single pulse

$$\begin{aligned} A(w) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv = \frac{1}{\pi} \int_{-1}^1 \cos wv \, dv = \frac{1}{\pi} \left. \frac{\sin wv}{w} \right|_{-1}^1 \\ &= \frac{2 \sin w}{\pi w} \end{aligned}$$

$$\begin{aligned} B(w) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv \, dv = \frac{1}{\pi} \int_{-1}^1 \sin wv \, dv = \frac{1}{\pi} \left. -\frac{\cos wv}{w} \right|_{-1}^1 \\ &= 0 \end{aligned}$$

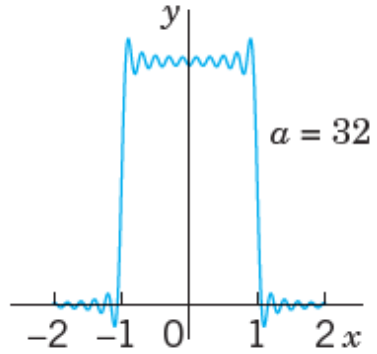
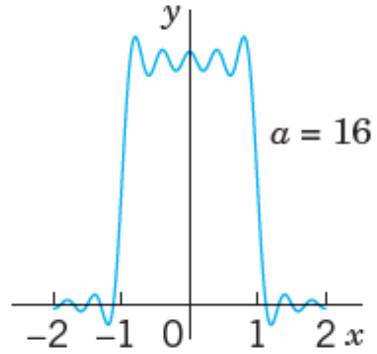
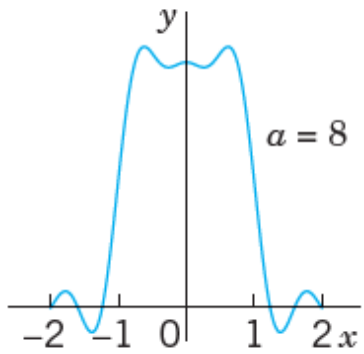
$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos wx \sin w}{w} dw \quad \text{Dirichlet's discontinuous factor}$$

$$\int_0^{\infty} \frac{\cos wx \sin w}{w} dw = \frac{\pi}{2} f(x) = \begin{cases} \frac{\pi}{2} & \text{if } 0 \leq x < 1 \\ \frac{\pi}{4} & \text{if } x = 1 \\ 0 & \text{if } x < 1 \end{cases}$$

Fourier Integral

Approximation

$$f(x) = \frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw$$



Gibbs phenomenon

Fourier Integral & Transforms

» Fourier cosine integral

$$f(-x) = f(x)$$

$$f(x) = \int_0^{\infty} A(w) \cos wx \, dw$$

$$A(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv$$

» Fourier cosine transforms

$$f_c(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx$$

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f_c(w) \cos wx \, dw$$

» Fourier sine integral

$$f(-x) = -f(x)$$

$$f(x) = \int_0^{\infty} B(w) \sin wx \, dw$$

$$B(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \sin wv \, dv$$

» Fourier sine transforms

$$f_s(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx$$

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f_s(w) \sin wx \, dw$$

Fourier Transforms

» Linearity theorem

$$(af + bg)_c = af_c + bg_c$$

$$(af + bg)_s = af_s + bg_s$$

» Transform of derivatives

- Let $f(x)$ be continuous and absolutely integrable on the x – axis, and let $f'(x)$ be piecewise continuous on every finite interval, and let $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$

$$f'_c = wf_s - \sqrt{\frac{2}{\pi}} f(0), \quad f''_c = -w^2 f_c - \sqrt{\frac{2}{\pi}} f'(0)$$

$$f'_s = -wf_c, \quad f''_s = -w^2 f_s + \sqrt{\frac{2}{\pi}} wf(0)$$

Fourier Transforms

$$\begin{aligned}
 f'_c = F_c(f'(x)) &= \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \cos wx \, dx \\
 &= \sqrt{\frac{2}{\pi}} \left[f(x) \cos wx \Big|_0^\infty + w \int_0^\infty f(x) \sin wx \, dx \right] \\
 &= \sqrt{\frac{2}{\pi}} [0 - f(0)] + wf_s \\
 f'_c &= wf_s - \sqrt{\frac{2}{\pi}} f(0)
 \end{aligned}$$

$$\begin{aligned}
 F_c(f''(x)) &= -\sqrt{\frac{2}{\pi}} f'(0) + wf'_s \\
 &= -\sqrt{\frac{2}{\pi}} f'(0) - w^2 f_c
 \end{aligned}$$

$$\begin{aligned}
 f'_s = F(f'(x)) &= \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \sin wx \, dx \\
 &= \sqrt{\frac{2}{\pi}} \left[f(x) \sin wx \Big|_0^\infty - w \int_0^\infty f(x) \cos wx \, dx \right] \\
 &= \sqrt{\frac{2}{\pi}} [(0 - 0) - wf_c] \\
 f'_s &= -wf_c
 \end{aligned}$$

$$\begin{aligned}
 F_s(f''(x)) &= -wf'_c = -w \left[-\sqrt{\frac{2}{\pi}} f(0) + wf'_s \right] \\
 &= w \sqrt{\frac{2}{\pi}} f(0) - wf_s
 \end{aligned}$$

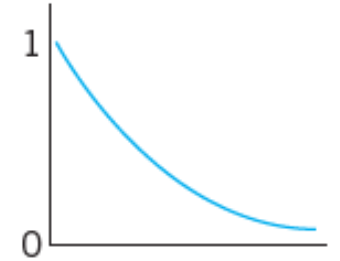
Practice Problems

Problem 1: Using Fourier cosine and sine integrals of $f(x)$

$$f(x) = e^{-kx}, \text{ where } x > 0 \text{ and } k > 0$$

Evaluate

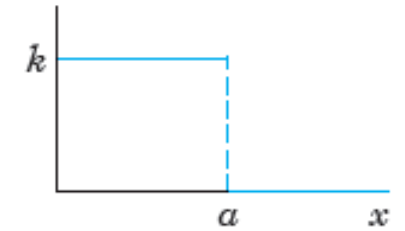
$$\int_0^{\infty} \frac{\cos wx}{k^2 + w^2} dw, \quad \int_0^{\infty} \frac{\sin wx}{k^2 + w^2} dw$$



$$f(x) = e^{-kx}$$

Problem 2: Find Fourier cosine and sine transform of $f(x)$

$$f(x) = \begin{cases} k, & \text{if } 0 < x < a \\ 0, & \text{if } x > a \end{cases}$$



Problem 3: Find Fourier cosine transform of $f(x)$

$$f(x) = e^{-ax}, \quad \text{where } a > 0$$

Fourier Integral

» Fourier integral: Complex form

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v) e^{iw(x-v)} dv dw$$

$$f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw$$

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv$$

Substitute $A(w)$ and $B(w)$ in (1), we get

$$\begin{aligned} f(x) &= \int_0^{\infty} \left[\left[\frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv \right] \cos wx + \left[\frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv \right] \sin wx \right] dw \\ &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(v) \cos wv \cos wx + \sin wv \sin wx dv dw \\ &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(v) \cos(wx - wv) dv dw \end{aligned}$$

$F(w)$ – this is an even function because $\cos(wx - wv)$ is an even function of w and $f(v)$ does not depend on w .

Fourier Integral

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v) \cos(wx - wv) dv dw \quad (1)$$

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v) \sin(wx - wv) dv dw = 0 \quad (2)$$

$$(1) + i \times (2)$$

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v) [\cos(wx - wv) + i \sin(wx - wv)] dv dw \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v) e^{i\omega(x-v)} dv d\omega \end{aligned} \quad e^{i\theta} = \cos \theta + i \sin \theta$$

Fourier Transform

» Fourier Transform: Complex form

- Let $f(x)$ be continuous and absolutely integrable on the x – axis, then

$$f(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx$$

The inverse Fourier transform is given by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(w) e^{iwx} dw$$

Fourier Transform

» Example

$$f(x) = \begin{cases} e^{-ax}, & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases}; \quad a > 0$$

$$f(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx$$

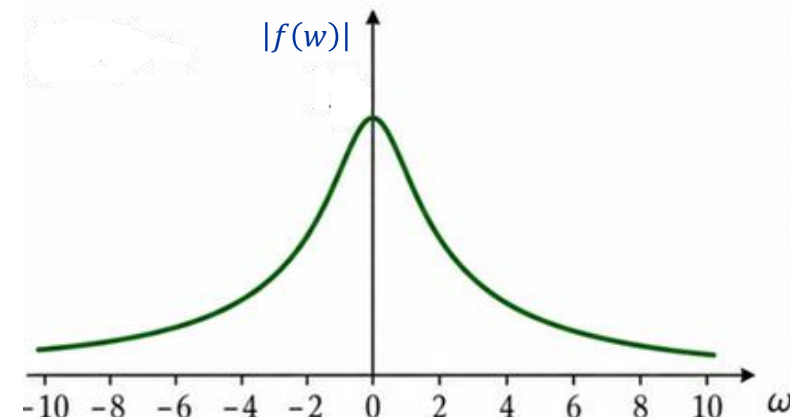
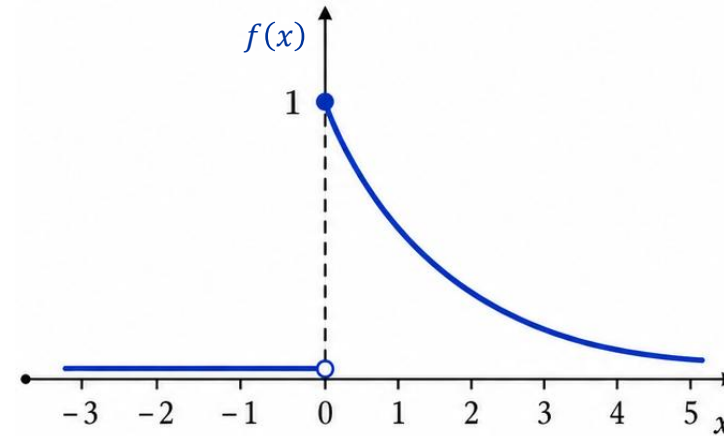
$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-ax} \cdot e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-(a+iw)x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left. \frac{e^{-(a+iw)x}}{-(a+iw)} \right|_0^{\infty}$$

$$= \frac{1}{\sqrt{2\pi}} \left(0 + \frac{1}{a+iw} \right)$$

$$f(w) = \frac{1}{\sqrt{2\pi}(a+iw)}$$



Fourier Transform

» Example

$$f(x) = e^{-a|x|}, \quad a > 0$$

$$f(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx$$

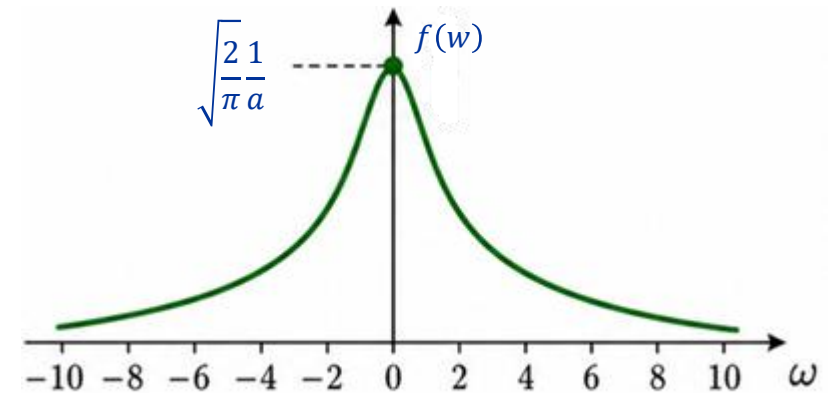
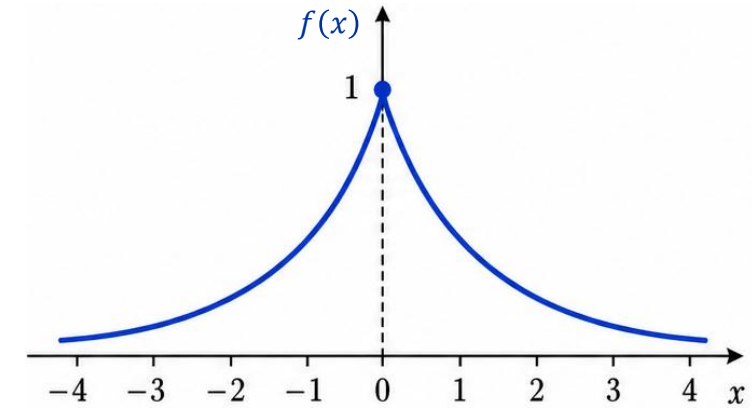
$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-a|x|} \cdot e^{-iwx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-a|x|} \cdot (\cos wx - i \sin wx) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-a|x|} \cos wx dx - \frac{i}{\sqrt{2\pi}} \int_0^{\infty} e^{-a|x|} \sin wx dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-ax} \cos wx dx$$

$$f(w) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2 + \omega^2}$$



Fourier Transform: Theorems

» Linearity theorem

$$\mathcal{F}(af + bg) = a\mathcal{F}(f) + b\mathcal{F}(g)$$

» Transform of derivatives

- Let $f(x)$ be continuous and absolutely integrable on the x – axis, and let $f'(x)$ be piecewise continuous on every finite interval, and let $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$

$$\mathcal{F}\{f'(x)\} = iw\{\mathcal{F}f(x)\}$$

$$\mathcal{F}\{f''(x)\} = -w^2\{\mathcal{F}f(x)\}$$

$$\begin{aligned}\mathcal{F}\{f'(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x)e^{-iwx}dx \\ &= \frac{1}{\sqrt{2\pi}} \left[f(x)e^{-iwx} \Big|_{-\infty}^{\infty} + iw \int_{-\infty}^{\infty} f(x)e^{-iwx}dx \right] \\ &= iw\{\mathcal{F}f(x)\}\end{aligned}$$

$$\begin{aligned}\mathcal{F}\{f''(x)\} &= iw\{\mathcal{F}f'(x)\} \\ &= (iw)iw\{\mathcal{F}f(x)\} \\ &= -w^2\{\mathcal{F}f(x)\}\end{aligned}$$

Fourier Transform: Theorems

» Frequency shifting theorem

$$\mathcal{F}(f(t)) = F(w)$$

$$\mathcal{F}(e^{w_0 t} f(t)) = F(w - w_0)$$

- Proof

$$\begin{aligned}\mathcal{F}(e^{w_0 t} f(t)) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{w_0 t} e^{-iwt} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i(w - w_0)t} dt \\ &= F(w - w_0)\end{aligned}$$

» Time shifting theorem

$$\mathcal{F}(f(t)) = F(w)$$

$$g(t) = \begin{cases} f(t - a), & \text{if } t > t_0 \\ 0, & \text{if } t < t_0 \end{cases}$$

$$\mathcal{F}(g(t)) = e^{-iwt_0} F(w)$$

- Proof

$$\begin{aligned}\mathcal{F}(g(t)) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(t) e^{-iwt} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{t_0}^{\infty} f(t) e^{-iwt} dt \\ (t - t_0) &= u, \quad dt = du \\ \mathcal{F}(g(t)) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(u) e^{-i w(t_0 + u)} du \\ &= e^{-iwt_0} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(u) e^{-i wu} du \\ \mathcal{F}(g(t)) &= e^{-iwt_0} F(w)\end{aligned}$$

Fourier Transform: Theorems

» Frequency shifting theorem

$$\mathcal{F}(f(t)) = F(\omega)$$

$$\mathcal{F}(f(\lambda t)) = \frac{1}{\lambda} F\left(\frac{\omega}{\lambda}\right), \quad \lambda > 0$$

- Proof

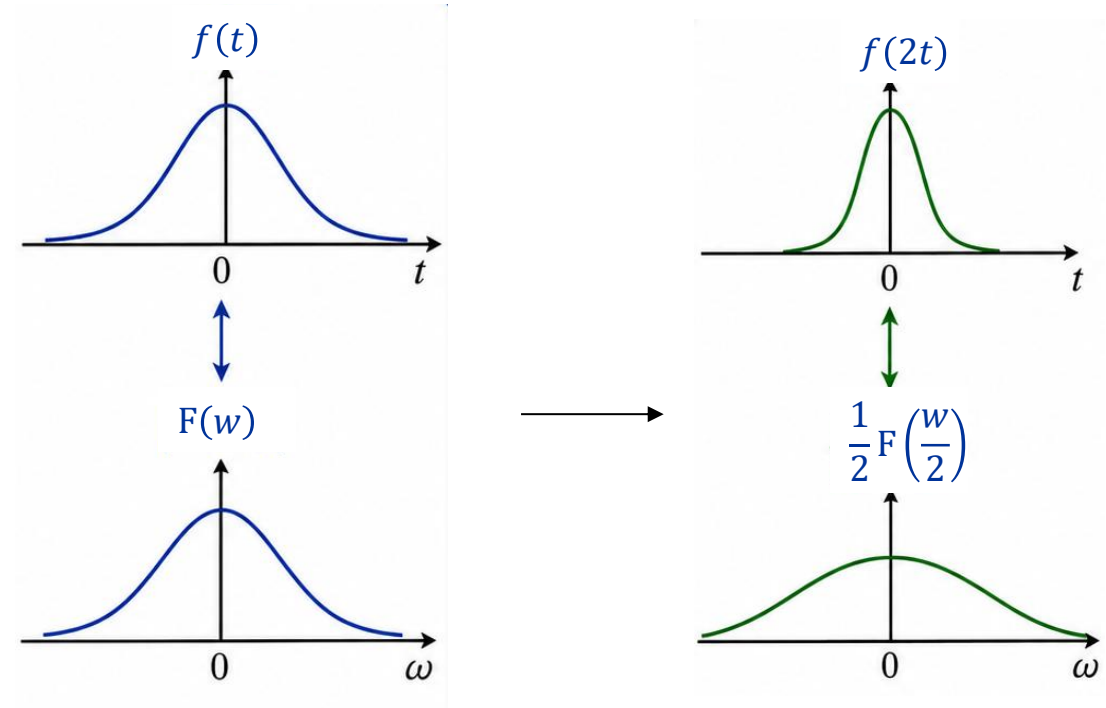
$$\mathcal{F}(f(\lambda t)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(\lambda t) e^{-i\omega t} dt$$

$$u = \lambda t, \quad du = \lambda dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{-i\frac{\omega}{\lambda}u} \frac{du}{\lambda}$$

$$= \frac{1}{\lambda} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{-i\frac{\omega}{\lambda}u} du$$

$$\mathcal{F}(f(\lambda t)) = \frac{1}{\lambda} F\left(\frac{\omega}{\lambda}\right), \quad \lambda > 0$$



Fourier Transform: Theorems

» Convolution

- Convolution $f * g$ of functions f and g is defined by

$$h(x) = (f * g)(x) = \int_{-\infty}^{\infty} f(p)g(x - p)dp = \int_{-\infty}^{\infty} f(x - p)g(p)dp$$

- Convolution theorem

$$\mathcal{F}(f * g) = \sqrt{2\pi} \mathcal{F}(f)\mathcal{F}(g)$$

$$(f * g)(x) = \int_{-\infty}^{\infty} \mathcal{F}(f)\mathcal{F}(g)e^{iwx}dw$$

$$\begin{aligned}\mathcal{F}(f * g) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x - p)g(p)dp e^{-iwx}dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x - p)g(p) e^{-iwx}dx dp\end{aligned}$$

Instead of x , we take $x - p = v$ as a new variable for integration.

$$x = p + v$$

Fourier Transform: Theorems

$$\begin{aligned}\mathcal{F}(f * g) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)g(v)e^{-i\omega(p+v)}dv dp \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(p)e^{-i\omega p}dp \cdot \int_{-\infty}^{\infty} g(v)e^{-i\omega v}dv \\ &= \frac{1}{\sqrt{2\pi}} \left(\sqrt{2\pi} \mathcal{F}(f) \right) \times \left(\sqrt{2\pi} \mathcal{F}(g) \right)\end{aligned}$$

$$\mathcal{F}(f * g) = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g)$$

Discrete Fourier Transform (DFT)

» Discrete Fourier transform

- Let $f(x)$ be periodic, for simplicity of period 2π . We assume N measurements of $f(x)$ are taken over the interval $0 \leq x \leq 2\pi$

$$x_k = \frac{2\pi k}{N}, \quad k = 0, 1, 2, \dots, N-1$$

Aim is to determine a complex trigonometric polynomial that interpolates $f(x)$ at the nodes

$$q(x) = \sum_{n=0}^{N-1} c_n e^{inx}$$

$$f_k = f(x_k) = q(x_k) = \sum_{n=0}^{N-1} c_n e^{inx_k}, \quad k = 0, 1, 2, \dots, N-1$$

The coefficient is given by

$$c_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-inx_k}, \quad n = 0, 1, 2, \dots, N-1$$

Discrete Fourier Transform (DFT)

$$f_k = \sum_{n=0}^{N-1} c_n e^{inx_k}, \quad k = 0, 1, 2, \dots, N-1$$

Multiply both sides of (1) by e^{-iwx} and sum over $k=0$ to $N-1$

$$\begin{aligned} \sum_{k=0}^{N-1} f_k e^{-imx} &= \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} c_n e^{i(n-m)x_k} \\ &= \sum_{n=0}^{N-1} c_n \sum_{k=0}^{N-1} e^{i(n-m)x_k} \\ &= \sum_{n=0}^{N-1} c_n \sum_{k=0}^{N-1} \underline{e^{i(n-m)\frac{2\pi k}{N}}} \end{aligned}$$

$$r = e^{i(n-m)\frac{2\pi}{N}}$$

For $n = m$; $r = e^0 = 1, c_m N$

Discrete Fourier Transform (DFT)

For $n \neq m$, we have $r \neq 1$

$$\sum_{k=0}^{N-1} r^k = (1 + r + r^2 + \dots + r^{N-1}) = \frac{1 - r^N}{1 - r}$$

$$r^N = e^{i(n-m)2\pi k} = \cos 2\pi k(n - m) + i \sin 2\pi k(n - m)$$

$$\frac{1 - r^N}{1 - r} = 0$$

$$\sum_{k=0}^{N-1} f_k e^{-imx} = c_m N$$

$$c_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-inx_k}, \quad n = 0, 1, 2, \dots, N-1$$

Discrete Fourier Transform (DFT)

» Discrete Fourier transform

- The discrete Fourier transform of the given signal, $\mathbf{f} = [f_0 \ \dots \ f_{N-1}]^T$

$$\hat{f}_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-inx_k}, \quad n = 0, 1, 2, \dots, N-1$$

- The DFT coefficient $\hat{f}_n$ is essentially the projection of the signal onto that frequency direction

In vector notation

$$\hat{\mathbf{f}} = \mathbf{F}_N \mathbf{f}$$

$\mathbf{F}_N = [e_{nk}]$ has the entries

$$e_{nk} = e^{-inx_k} = e^{-2\pi ink/N} = w^{nk}, w = w_N = e^{-2\pi i/N}$$

where $n, k = 0, 1, 2, \dots, N-1$

$$\mathbf{f} = \mathbf{F}_N^{-1} \hat{\mathbf{f}} = \frac{1}{N} \overline{\mathbf{F}_N} \hat{\mathbf{f}}$$

($\overline{\mathbf{F}_N}$ is complex conjugate of $\mathbf{F}_N$)

» Example

Let $N = 4$, and $\mathbf{f} = [0 \ 1 \ 4 \ 9]^T$. Find DFT of $\mathbf{f}$?

$$\hat{\mathbf{f}} = \mathbf{F}_4 \mathbf{f}$$

$$w = e^{-2\pi i/N} = e^{-2\pi i/4} = e^{-i\pi/2} = -i$$

$$w^{nk} = (-i)^{nk}; \quad \begin{matrix} n = 0, 1, 2, 3 \\ k = 0, 1, 2, 3 \end{matrix}$$

$$\mathbf{F}_4 = \begin{bmatrix} w^0 & w^0 & w^0 & w^0 \\ w^0 & w^1 & w^2 & w^3 \\ w^0 & w^2 & w^4 & w^6 \\ w^0 & w^3 & w^6 & w^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

$$\therefore \hat{\mathbf{f}} = \mathbf{F}_4 \mathbf{f} = \begin{bmatrix} 14 \\ -4 + 8i \\ -6 \\ -4 - 8i \end{bmatrix}$$

Discrete Fourier Transform (DFT)

» Fast Fourier transform (FFT)

DFT: Computational cost $\sim \mathbf{O(N^2)}$

FFT: Computational cost $\sim \mathbf{O(N) \log_2 N}$

Break the problem into two problems of size $M = N/2$

$$f = [f_0 \ f_1 \ f_2 \ \dots \ f_{N-1}]^T$$

$$f_{\text{even}} = [f_0 \ f_2 \ \dots \ f_{N-2}]^T ; \quad f_{\text{odd}} = [f_1 \ f_3 \ \dots \ f_{N-1}]^T$$

$$\therefore \hat{f}_{\text{even}} = F_M f_{\text{even}} ; \quad \hat{f}_{\text{odd}} = F_M f_{\text{odd}}$$

$$\hat{f}_n = \hat{f}_{\text{even},n} + w_N^n \hat{f}_{\text{odd},n} ; \quad n = 0, 1, 2, \dots, M-1$$

$$\hat{f}_{n+M} = \hat{f}_{\text{even},n} - w_N^n \hat{f}_{\text{odd},n} ; \quad n = 0, 1, 2, \dots, M-1$$

Discrete Fourier Transform (DFT)

» Example

Let $N = 4$, and $f = [0 \ 1 \ 4 \ 9]^T$. Find DFT of f ?

$$f_{\text{even}} = \begin{bmatrix} f_0 \\ f_2 \end{bmatrix} ; f_{\text{odd}} = \begin{bmatrix} f_1 \\ f_3 \end{bmatrix} ; M = \frac{N}{2} = 2$$

$$\hat{f}_{\text{even}} = F_M f_{\text{even}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_0 \\ f_2 \end{bmatrix} = \begin{bmatrix} f_0 + f_2 \\ f_0 - f_2 \end{bmatrix} \quad \hat{f}_{\text{odd}} = F_M f_{\text{odd}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_3 \end{bmatrix} = \begin{bmatrix} f_1 + f_3 \\ f_1 - f_3 \end{bmatrix}$$

$$\hat{f}_0 = \hat{f}_{\text{even},0} + w_N^0 \hat{f}_{\text{odd},0} = (f_0 + f_2) + (f_1 + f_3) = f_0 + f_1 + f_2 + f_3$$

$$\hat{f}_1 = \hat{f}_{\text{even},1} + w_N^1 \hat{f}_{\text{odd},1} = (f_0 - f_2) - i(f_1 + f_3) = f_0 - if_1 - f_2 + if_3$$

$$\hat{f}_2 = \hat{f}_{\text{even},0} - w_N^0 \hat{f}_{\text{odd},0} = (f_0 + f_2) - (f_1 + f_3) = f_0 - f_1 + f_2 - f_3$$

$$\hat{f}_3 = \hat{f}_{\text{even},1} - w_N^1 \hat{f}_{\text{odd},1} = (f_0 - f_2) - (-i)(f_1 + f_3) = f_0 + if_1 - f_2 - if_3$$

Discrete Fourier Transform (DFT)

» MATLAB

$$f(x) = \sin x + \sin(3x)$$

% Sampling

```
N = 8;  
n = 0:N-1;  
x = 2*pi*n/N;  
f = (sin(x) + sin(3*x))';
```

% Root of unity

```
w = exp(-1i*2*pi/N);
```

% Fourier matrix

```
F = zeros(N,N);
```

```
for k = 0:N-1  
    for n = 0:N-1  
        F(k+1,n+1) = w^(k*n);  
    end  
end
```

% Forward DFT

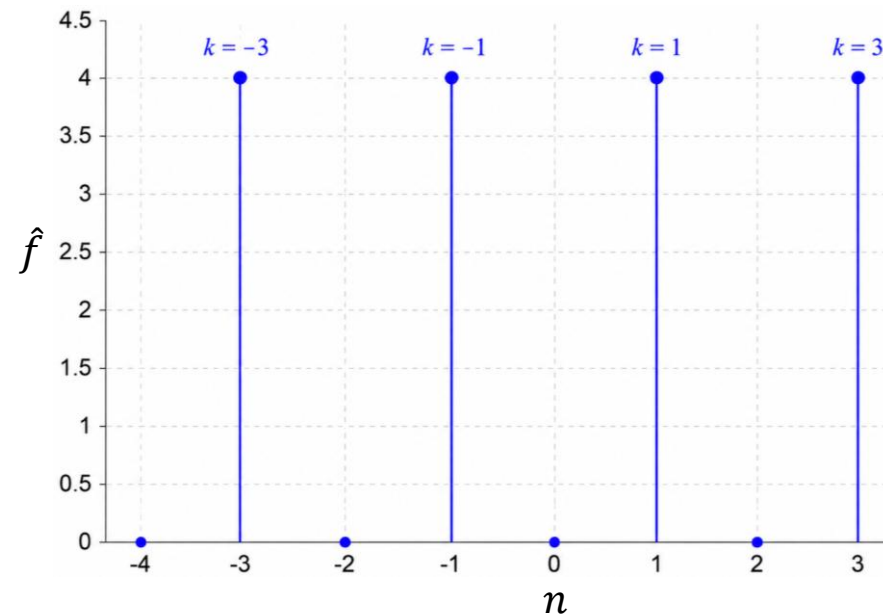
```
f_hat = F*f;
```

% Inverse DFT

```
f = (1/N)*conj(F)*Fhat;
```

% DFT using MATLAB fft

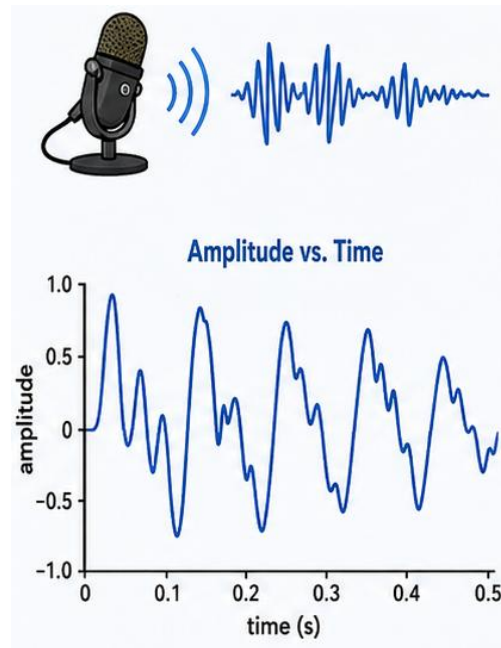
```
F_hat = fft(f);
```



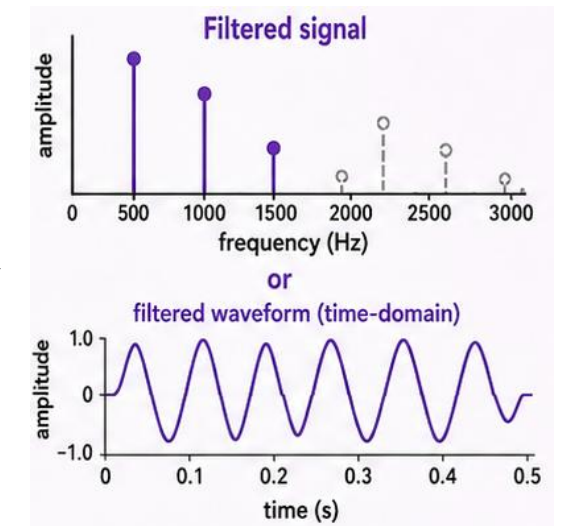
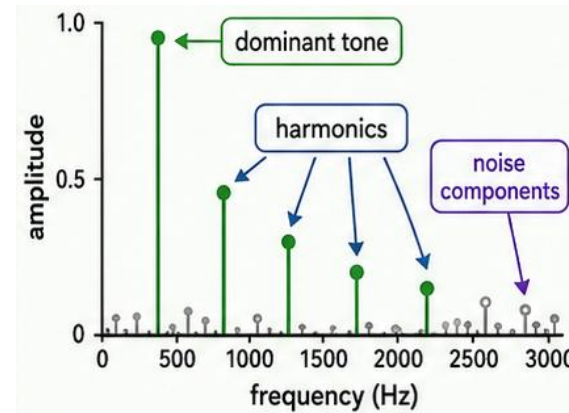
Applications

» Application: sound analysis

- Help separate mixed frequencies
- Reduce noise
- Design audio filters



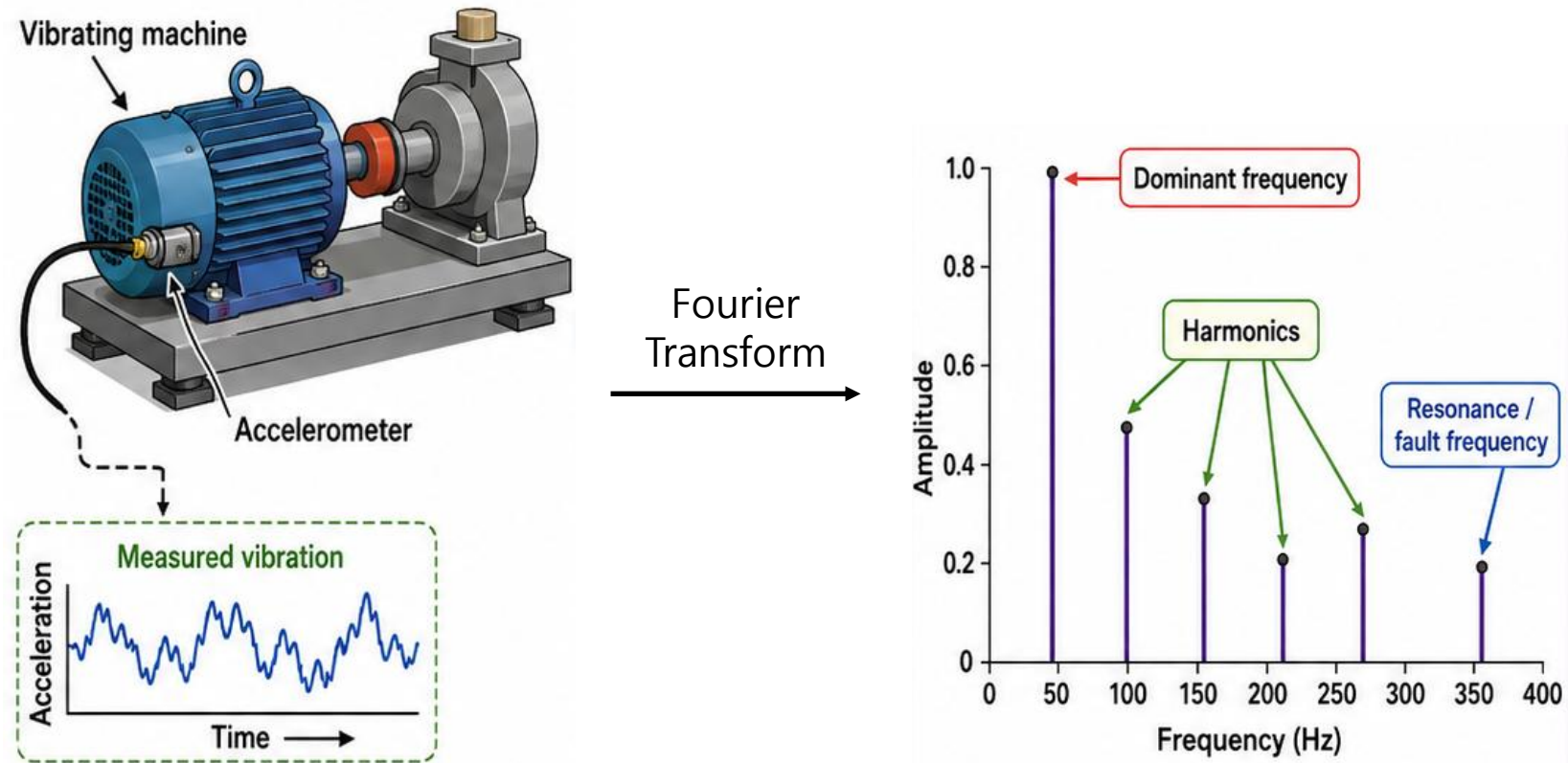
Fourier
Transform



Applications

» Application: vibration analysis

- Identify natural frequencies
- Detect imbalance, misalignment, and bearing faults
- Monitor machine health



Practice Problems

Problem 1: Find the Fourier transform of $f(x)$

$$f(x) = xe^{-x^2}$$



Problem 2: Find $f(x)$, if Fourier transform is

$$\frac{e^{-aw}}{w}$$